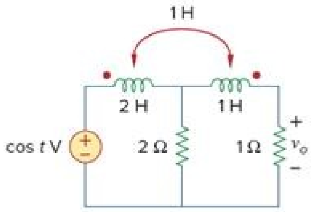
# Chapter 18, Problem 49P

## Problem

Use the Fourier transform method to obtain *vo*(*t*) in the circuit of Fig. 18.51.

**Figure 18.51**



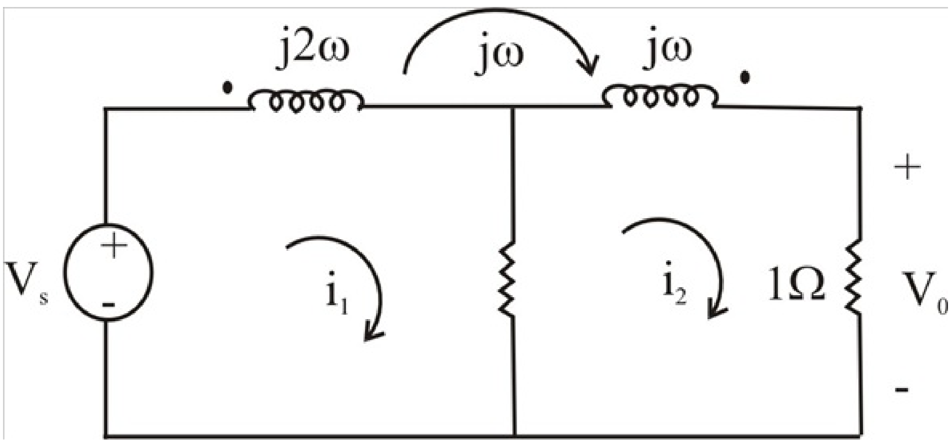

## Step-by-step solution

**Step 1** of 4

Consider the circuit shown below:-

$$V_s = \pi\left[s(\omega+1)+s(\omega-2)\right]$$

**Step 2** of 4




**Step 3** of 4

$$\begin{aligned}
&\textit{for mesh } 1, && -V_s+(2+j2\omega)I_1-2I_2-j\omega I_2=0\\
&\textit{for mesh } 2, && 2(1+j\omega)I_2-(2+j\omega)I_2 \qquad -(1)\\
&\textit{for mesh } 3, && 0=(3+j\omega)I_2-2I_1-j\omega I_1\\
& && I_1=\frac{(3+\omega)I_2}{(2+\omega)} \qquad -(2)
\end{aligned}$$

**Step 4** of 4

Subsituting $(2)$ into $(1)$ gives

$$V_s = 2\frac{2(1+j\omega)(3+j\omega)}{(2+j\omega)}I_2 - (2+j\omega)I_2$$

$$V_s(2+j\omega) = \left[2\left(3+j4\omega-\omega^2\right)-\left(4+j4\omega-\omega^2\right)\right]I_2$$

$$I_2 = \frac{(s+2)V_s}{s^2+4s+2}, \qquad s = j\omega$$

$$V_o = I_2\frac{(j\omega+2)\pi\left[s(\omega+1)+s(\omega-1)\right]}{(j\omega)^2`+4j\omega+2}$$

$$V_o(t) = \frac{1}{2\pi}\int_{-\infty}^{0} V_o(\omega)e^{j\omega t}d\omega$$

$$= \int_{-\infty}^{\infty} \frac{\frac{1}{2}(j\omega+2)e^{j\omega t}s(\omega+1)d\omega}{(j\omega)^2+4j\omega+2} + \frac{\frac{1}{2}(j\omega+2)e^{j\omega t}s(\omega-1)d\omega}{(j\omega)^2+4j\omega+2}$$

$$= \frac{\frac{1}{2}(-j+2)e^{jt}}{-1-4j+2} + \frac{\frac{1}{2}(j+2)e^{jt}}{-1+j4+2}$$

$$V_o(t) = \frac{\frac{1}{2}(2+j)(1+4j)e^{jt}}{17} + \frac{\frac{1}{2}(2-j)(1-4j)e^{jt}}{17}$$

$$= \frac{1}{34}(6+j7)e^{jt} + \frac{1}{34}(6-7j)e^{jt}$$

$$v_o(t) = 0.542\cos\left(t-13.64^o\right)V$$